Linking Sublethal Copper Toxicity Exposure to Respiration Response in Monogonont Rotifers

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Abstract:
Anthropogenic metal contamination challenges freshwater ecosystems, and trace concentrations can impair physiological functions of organisms. Rotifers are valuable models for assessing heavy metal toxicity in aquatic habitats. We evaluated whether long-term habitat history influences tolerance to copper sulfate exposure by comparing two populations each of Epiphanes brachionus and E. chihuahuaensis, isolated from polluted and non-polluted temporary environments. Acute toxicity tests were used to determine lethal concentrations (LC50) of 0.041 mg/L for E. brachionus and 0.068 mg/L for E. chihuahuaensis, both below the mean for monogonont rotifers (~0.096 mg/L). To determine sublethal effects, rotifers were exposed to 70% of the LC50 for 24 h, followed by a 2 h recovery period, during which oxygen consumption was measured. Control respiration rates for neonates averaged 0.68 and 1.2 pmol oxygen/min/ind for E. brachionus and E. chihuahuaensis, respectively; adult rates were ~ 4 times higher. Following copper exposure, neonate E. brachionus from a polluted habitat exhibited a 20.6% decrease in respiration relative to controls, whereas E. chihuahuaensis from a non-polluted habitat showed an 82.4% increase. These preliminary findings show lineage- and history-specific metabolic effects of long-term exposure to pollutants. This study links mortality thresholds and physiological endpoints such as respiration within historical exposure context in ephemeral pools, advancing our understanding of toxicant stress response and reiterates the value of rotifers as bioindicators.

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